Project 6 Report: Weekly Radar

Anomaly Detection on Weekly Network Performance

Carrier and Region Trend Analysis for Egyptian Governorates

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1 Introduction

This report documents the final corrected workflow for **Project 6 — Weekly Radar** from the ITI Telco-Cloud Analytics training. The project analyzes weekly mobile network performance across Egyptian governorates, carriers, and radio technologies.

The workflow has two main notebook phases:

1. a **preprocessing notebook** that cleaned the original workbook, separated weekly locality rows from country aggregates, and built the final region-carrier-technology-week table; and
2. an **analysis notebook** that applied cross-sectional peer comparison, week-over-week change analysis, LTE-vs-5G comparison, regional ranking, and mean-vs-median interpretation.

The key analytical challenge is that the weekly dataset contains only three weekly snapshots: 2026-03-01, 2026-03-08, and 2026-03-15. Therefore, the project is not a long-history time-series anomaly problem. The strongest approach is to compare entities that exist at the same time: carriers against same-region peers, regions against other regions, and technologies across matched region-week records.

2 Project Objective and Analytical Design

The project brief asks five main questions:

Q1. Which carriers are consistently below the regional average for download speed across all three weeks?

Q2. Which regions show the largest week-over-week change in download speed or latency?

Q3. Are there region-carrier pairs that are outliers inside their region?

Q4. Is the 5G performance distribution different from LTE across regions?

Q5. How do the 13 regions rank from best to worst based on median download speed, and which region has the most unstable rank?

The final design used the following principles:

- **Weighted aggregation:** KPI values were aggregated using `sample_count` as reliability weight.
- **Median proxy:** regional median KPI columns are sample-count-weighted averages of locality medians. They are strong proxies, but not exact combined medians because raw speed-test observations are not available.

- **Reliability threshold:** the final preprocessing threshold was reduced from 50 to `sample_count_total >= 30` to preserve full region-week coverage while still filtering weak records.
- **Complete peer groups:** Q1 and Q3 use only region-week-technology groups where all three carriers exist, making peer comparisons fair.
- **Interpretability:** every anomaly score is tied back to a named region, carrier, technology, week, and KPI direction.

3 Preprocessing Notebook

3.1 Why preprocessing was necessary

The original spreadsheet mixed daily and weekly rows, locality and country rows, and many repeated locality records that had to be aggregated into a clean project-level unit. The final analysis unit is:

$$\text{region} \times \text{carrier} \times \text{technology} \times \text{week}$$

Processing stage	Rows
Raw spreadsheet rows	93,364
Weekly rows kept	58,858
Daily rows removed	34,506
Weekly locality rows	58,669
Weekly country rows	189
Aggregated region-carrier-technology-week rows	351
Rows removed by sample threshold	8
Final analysis rows	343

Table 1: Row counts through the corrected preprocessing pipeline.

Table 1 shows the corrected pipeline. The workbook started with 93,364 rows. After weekly filtering, 58,858 rows remained. Locality rows were used for the main region-level analysis, while 189 Egypt-level country aggregates were separated as optional reference rows. After region-carrier-technology-week aggregation, 351 rows existed before reliability filtering. The threshold `sample_count_total >= 30` removed only 8 weak rows and left 343 final rows.

3.2 Date parsing and validation

One important correction was the date parser. The final notebooks avoid forcing `dayfirst=True` on already-standard ISO dates such as `2026-03-15`. This prevents

the third week from being converted incorrectly or dropped. The corrected validation confirms that all three weekly snapshots are present.

Validation check	Final result
Main cleaned table	343 region-carrier-technology-week rows
Weeks present	3 / 3
Regions present	13 / 13
Region-week coverage	39 / 39
Region-technology-week coverage	117 / 117
Region-carrier-week coverage	117 / 117
Rows used for complete peer comparison	330
Rows excluded from peer comparison	13

Table 2: Final coverage checks after applying the threshold of `sample_count_total >= 30`.

3.3 Why aggregation and weighting were required

The locality-level rows are not strong enough individually. Many locality rows contain very small sample counts, while the aggregated regional records are much more reliable. Therefore, each KPI was aggregated with:

$$\bar{x}_w = \frac{\sum_i x_i \cdot n_i}{\sum_i n_i}$$

where x_i is the KPI value and n_i is the `sample_count` behind that locality record.

Dataset	Median	Mean	Min	Max
Weekly locality rows	2	23.00	1	6,732
Aggregated rows before threshold	972	3,844.27	8	38,163
Final rows after threshold	1,009	3,933.50	30	38,163

Table 3: Sample-count quality before and after region-level aggregation.

The locality-level weekly rows had a median sample count of only 2, while the final filtered region-level rows had a median sample count of 1,009. This is why the weighted aggregation step is not just a coding detail; it is part of the reliability design.

4 Analysis Notebook: Methods

4.1 Threshold audit

The final notebooks use a small number of threshold-like decisions. Table 4 summarizes each one and why it is appropriate.

Decision	Value	Reason
Preprocessing sample threshold	`sample_count_total >= 30`	Keeps all 39 region-week combinations while removing only the weakest 8 aggregated rows.
Complete peer-group rule	exactly 3 carriers	Ensures Q1 and Q3 compare carriers only against complete same-region, same-week, same-technology peer groups.
Peer anomaly cutoff	95th percentile = 84.89	Keeps the most severe cross-sectional cases from 330 comparable rows.
Week-over-week anomaly cutoff	90th percentile = 1.319	Less brittle than 95th percentile because only 26 region transition rows exist.
Q2 reporting list	`TOP_N = 5`	Readable reporting rule for largest download drops and latency rises; not a statistical anomaly threshold.

Table 4: Threshold audit used in the final preprocessing and analysis notebooks.

4.2 Method 1: Cross-sectional peer comparison

For each `region` × `week` × `technology` group, each carrier was compared against the weighted average of the other carriers in that same peer group.

$$\text{Download Gap (\%)} = \frac{\text{Carrier Download} - \text{Peer Average Download}}{\text{Peer Average Download}} \times 100$$

$$\text{Latency Gap (\%)} = \frac{\text{Carrier Latency} - \text{Peer Average Latency}}{\text{Peer Average Latency}} \times 100$$

A negative download gap is bad because the carrier is slower than peers. A positive latency gap is bad because the carrier is more latent than peers. The final peer anomaly score follows the same formula used in the slides:

$$\text{Cross-sectional Badness} = \max(0, -\text{Download Gap (\%)}) + \max(0, \text{Latency Gap (\%)})$$

The score is measured in combined bad percentage-gap points. For example, a row with download 60% below peers and latency 30% above peers receives a score of 90.

4.3 Method 2: Week-over-week regional change analysis

The second method identifies regional changes between consecutive weeks. Carrier and technology rows were combined into one weighted regional summary per week. Then the following equation was applied:

$$\text{WoW Change (\%)} = \frac{\text{Current Week KPI} - \text{Previous Week KPI}}{\text{Previous Week KPI}} \times 100$$

Large negative download changes and large positive latency changes were treated as deterioration signals. The analysis also keeps direct-answer tables for the largest download drops and the largest latency rises, because a region can deteriorate in only one KPI.

5 Results and Discussion

5.1 Q1: Which carriers are consistently below the regional average?

The corrected Q1 result is more nuanced than the first report version. **Operator C** is still the main consistent underperformer because it has the strongest negative download gap, the highest weighted below-peer rate, and 16 of the 17 severe peer anomaly flags. **Operator B** is slightly below the weighted peer baseline across the three weeks, but its gap is small and almost neutral by 2026-03-15. **Operator A** remains clearly above peers overall.

Carrier	2026-03-01 gap	2026-03-08 gap	2026-03-15 gap	Overall gap	Below-peer rate	Flags	Interpretation
Operator A	29.94	28.66	27.08	28.57	48.0%	0	Above peers overall
Operator B	-5.54	-4.73	-0.06	-3.54	50.6%	1	Slightly below; near neutral by week 3
Operator C	-12.13	-6.92	-13.58	-10.81	63.5%	16	Main consistent underperformer

Table 5: Carrier download gap relative to the weighted regional peer baseline. Gaps are percentages; negative values mean below peers.

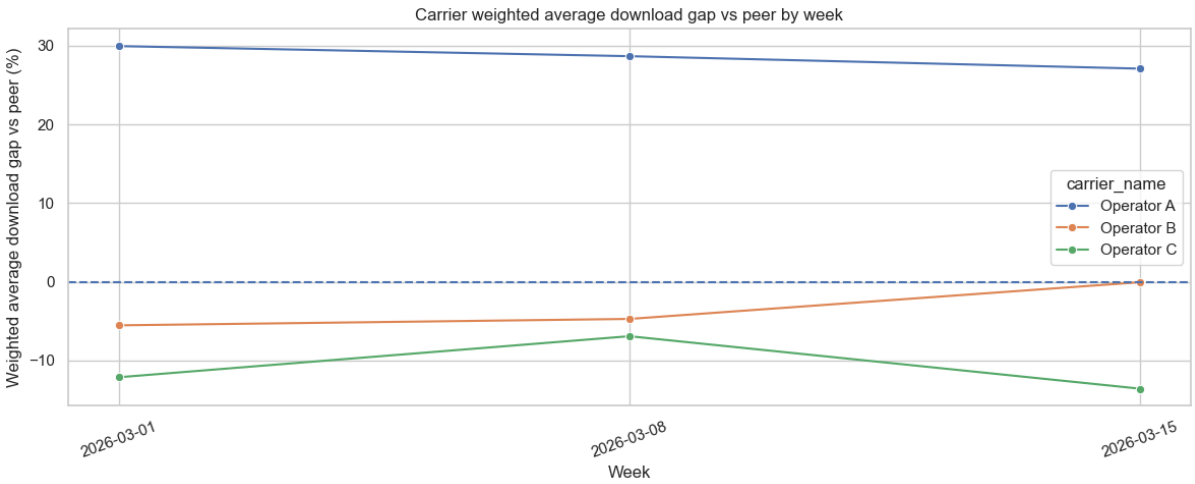

Figure 1: Weighted average carrier download gap relative to regional peers by week.

5.2 Q2: Which regions had the largest week-over-week changes?

The largest **download deterioration** cases are shown in Table 6. These are direct download drops, not necessarily combined download-and-latency anomaly flags.

Week	Region	Download change (%)	Latency change (%)
2026-03-15	Assiut	-14.54	-0.04
2026-03-15	Monufia	-13.31	1.27
2026-03-15	Gharbia	-13.29	2.19
2026-03-15	Minya	-12.96	2.22
2026-03-08	Alexandria	-12.93	-1.29

Table 6: Largest negative week-over-week download changes.

The largest **latency deterioration** cases are shown in Table 7.

Week	Region	Download change (%)	Latency change (%)
2026-03-08	Monufia	9.78	4.85
2026-03-15	Sohag	9.55	4.73
2026-03-15	Dakahlia	-6.33	4.08
2026-03-15	Fayoum	9.74	3.24
2026-03-08	Sharqia	-0.15	3.10

Table 7: Largest positive week-over-week latency changes.

The combined week-over-week anomaly score is useful when both directions are considered together. The top combined cases are shown in Table 8 and Figure 2. Dakahlia appears twice because it had both download decline and latency increase across consecutive transitions.

Week	Region	Download change (%)	Latency change (%)	Combined score	Flagged?
2026-03-08	Dakahlia	-10.98	2.86	1.611	Yes
2026-03-15	Dakahlia	-6.33	4.08	1.389	Yes
2026-03-08	Port Said	-7.34	2.70	1.333	Yes
2026-03-15	Gharbia	-13.29	2.19	1.306	No
2026-03-15	Minya	-12.96	2.22	1.292	No

Table 8: Top combined week-over-week anomaly scores.

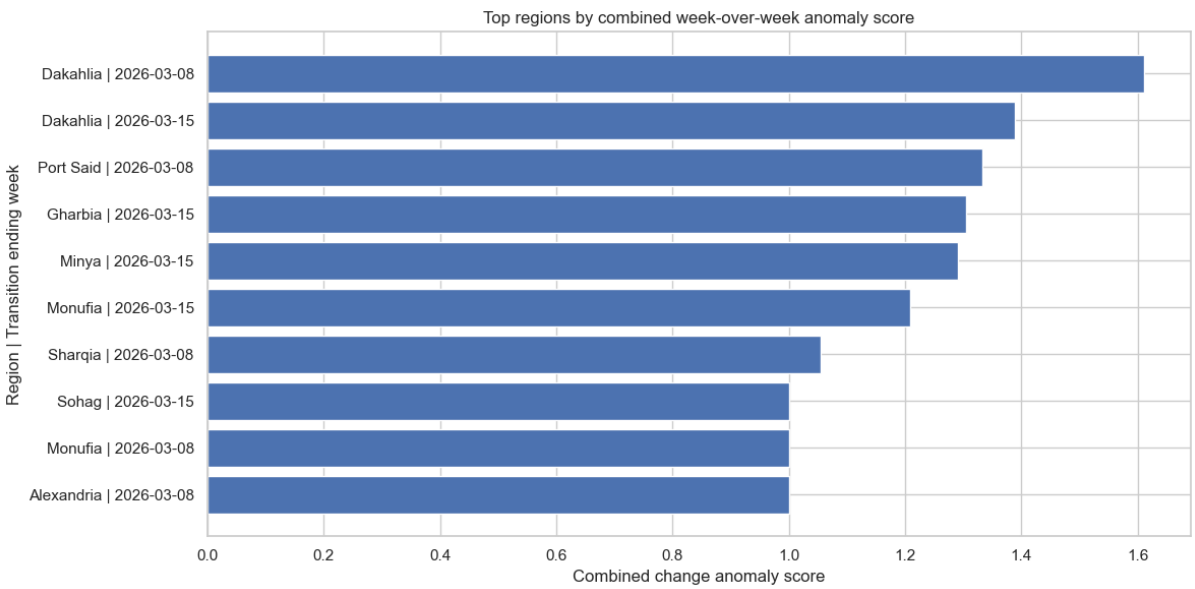

Figure 2: Top regions by combined week-over-week anomaly score.

5.3 Q3: Which region-carrier pairs were outliers within their region?

The strongest peer anomalies are dominated by **Operator C on LTE**, especially in Assiut, Ismailia, Giza, Alexandria, and Gharbia. These cases combine two negative signs: download below peers and latency above peers.

Week	Region	Carrier	Tech	Download gap (%)	Latency gap (%)	Score
2026-03-08	Assiut	Operator C	LTE	-67.46	70.65	138.11
2026-03-01	Assiut	Operator C	LTE	-52.03	83.96	135.99
2026-03-15	Assiut	Operator C	LTE	-68.50	66.79	135.29
2026-03-08	Ismailia	Operator C	LTE	-61.68	69.12	130.79
2026-03-01	Ismailia	Operator C	LTE	-61.89	64.37	126.26
2026-03-01	Alexandria	Operator C	LTE	-65.18	57.86	123.04
2026-03-15	Gharbia	Operator C	LTE	-56.64	63.21	119.86
2026-03-15	Ismailia	Operator C	LTE	-51.13	57.46	108.58

Table 9: Top cross-sectional peer anomalies after applying the slide badness formula.

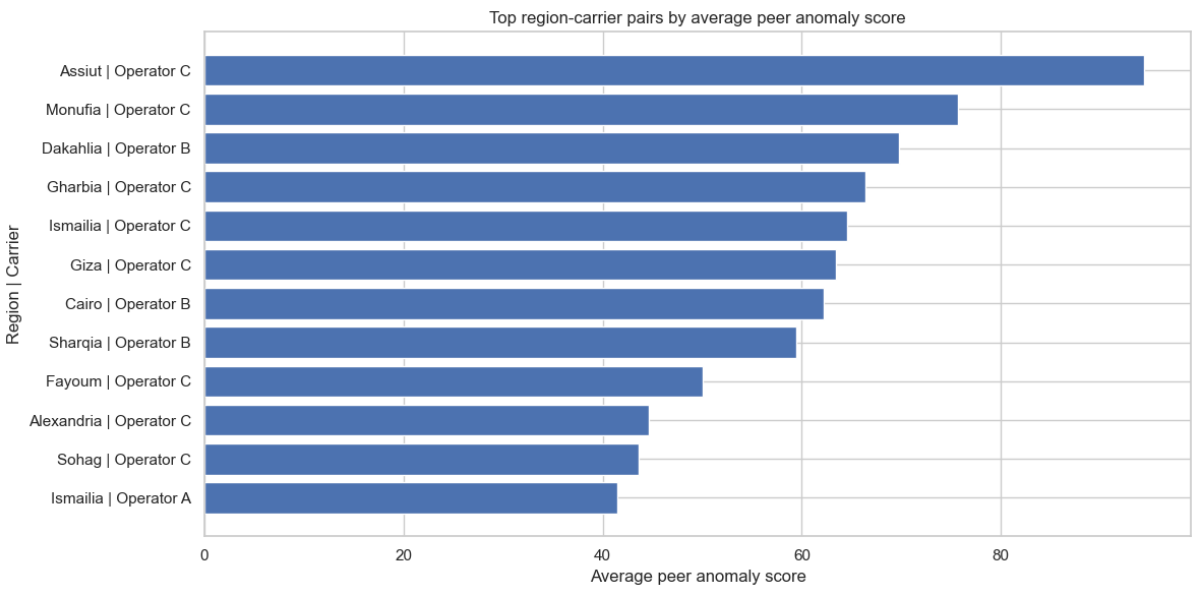

Figure 3: Top region-carrier pairs by average peer anomaly score.

5.4 Q4: Is the 5G distribution different from LTE?

Yes. 5G is substantially stronger than LTE across the final cleaned data.

Technology	Observations	Avg download (kbps)	Median download (kbps)	Avg latency (ms)
5G	39	295,221	301,663	31.30
LTE	39	48,988	50,830	38.02
Multi-RAT	39	105,193	111,827	35.83

Table 10: Distribution summary by technology across all 39 region-week observations.

Across all 39 region-week comparisons, 5G never underperformed LTE in median download. The average 5G-minus-LTE gap was approximately 246,233 kbps. A stricter matched-carrier check was also performed using 109 region-week-carrier comparisons where both LTE and 5G existed. In those matched cases, 5G underperformed LTE in **0** cases.

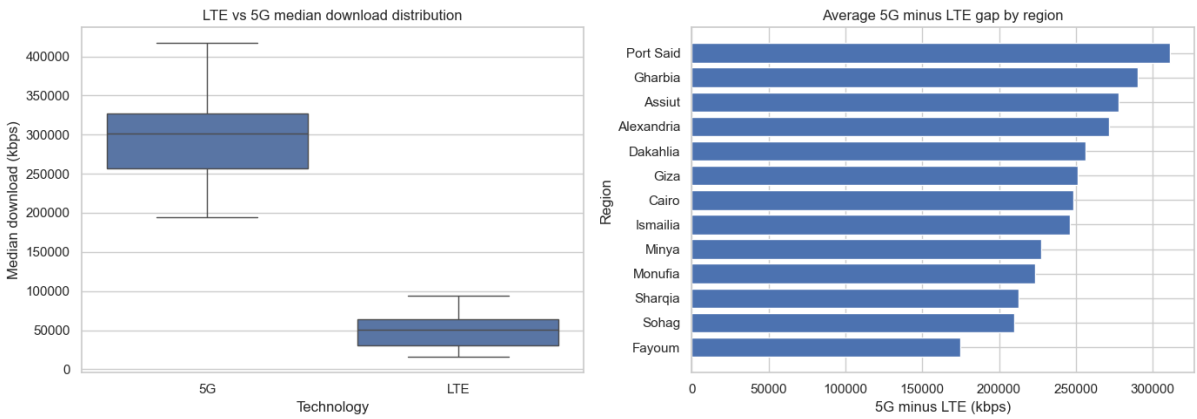

Figure 4: LTE vs 5G median download distribution and average 5G-minus-LTE gap by region.

5.5 Q5: Regional ranking and rank stability

The regional ranking was built from the weighted median-download proxy after combining carriers and technologies using sample-count weights. Rank 1 is the best.

Region	2026-03-01	2026-03-08	2026-03-15
Alexandria	4	5	6
Assiut	3	2	3
Cairo	10	9	8
Dakahlia	9	10	10
Fayoum	12	11	11
Gharbia	2	1	2
Giza	5	4	4
Ismailia	6	6	5
Minya	7	8	9
Monufia	8	7	7
Port Said	1	3	1
Sharqia	11	12	12
Sohag	13	13	13

Table 11: Regional ranking based on aggregated median download speed proxy (1 = best).

The strongest regions were generally **Port Said**, **Gharbia**, and **Assiut**. The weakest were consistently **Sohag**, **Sharqia**, and **Fayoum**. Sohag remained rank 13 in all three weeks, which makes it weak but stable. The most unstable region was **Port Said**, whose rank moved from 1 to 3 and back to 1.

Region	Rank range	Total absolute rank movement
Port Said	2	4
Alexandria	2	2
Cairo	2	2
Minya	2	2
Assiut	1	2

Table 12: Most unstable regions by ranking movement.

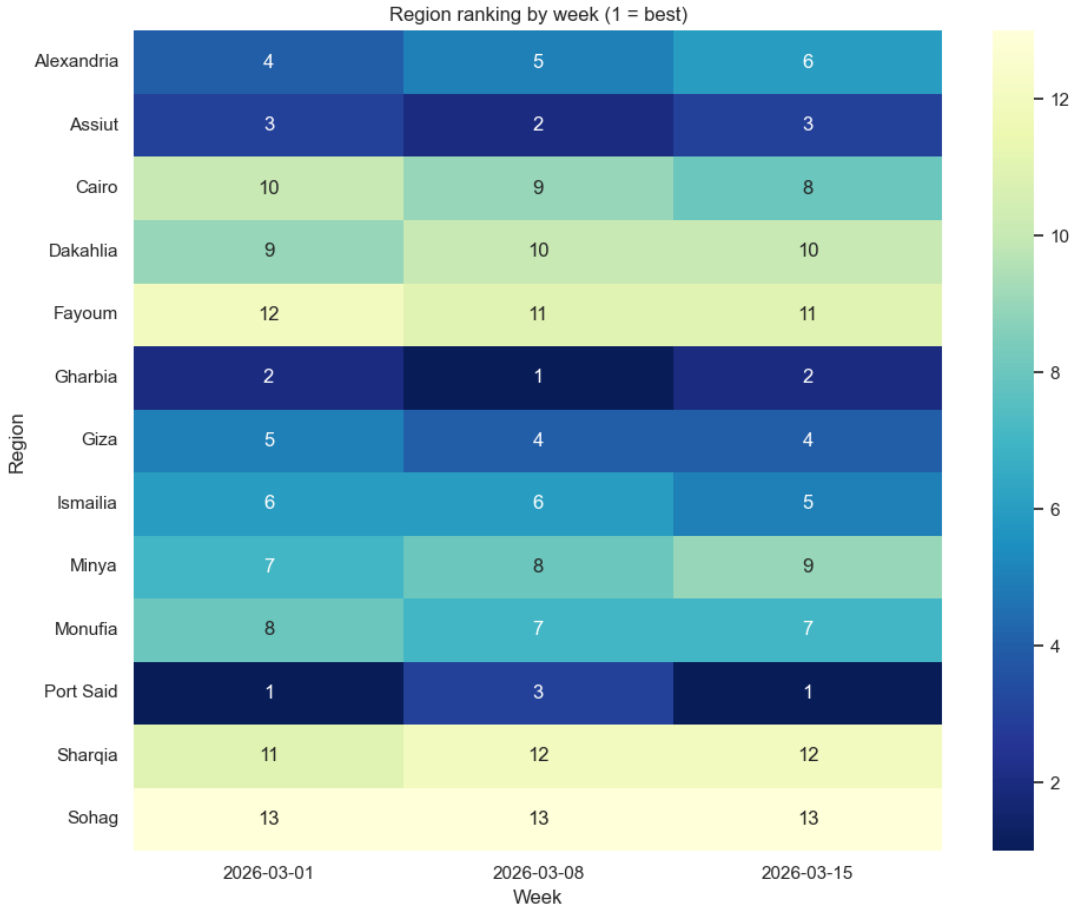

Figure 5: Regional rank trajectories across the three weekly snapshots (1 = best).

6 Additional Required Deliverables

6.1 Carrier report

The carrier report compares Operators A, B, and C across regions and weeks. The final version separates two different meanings of "worst carrier":

- **Worst by download:** the carrier with the lowest average weighted median-download proxy in the region.
- **Worst by peer anomaly score:** the carrier with the highest average cross-sectional badness score in the region.

This separation is important because a carrier can have the lowest absolute download in a region without being the largest relative peer anomaly.

Region	Worst by download	Worst by peer score	Avg peer score
Alexandria	Operator A	Operator C	44.66
Assiut	Operator A	Operator C	94.35
Cairo	Operator B	Operator B	62.24
Dakahlia	Operator B	Operator B	69.73
Fayoum	Operator C	Operator C	50.09
Gharbia	Operator A	Operator C	66.41
Giza	Operator A	Operator C	63.47
Ismailia	Operator A	Operator C	64.51
Minya	Operator A	Operator A	20.51
Monufia	Operator C	Operator C	75.70
Port Said	Operator A	Operator B	17.37
Sharqia	Operator B	Operator B	59.46
Sohag	Operator C	Operator C	43.60

Table 13: Worst carrier per region under two definitions.

6.2 Anomaly logs

Three CSV deliverables were generated for reproducibility and presentation use:

1. `project6_peer_anomaly_log.csv`
2. `project6_change_anomaly_log.csv`
3. `project6_anomaly_log.csv` (combined log)

The combined anomaly log contains 20 flagged rows: 17 cross-sectional peer anomalies and 3 week-over-week change anomalies.

6.3 Mean vs median analysis

Project 6 asks for an explanation of where mean and median diverge. Large positive gaps between mean and median often suggest a right-skewed distribution, where a few unusually high values pull the mean upward. For latency, a large mean above the median suggests a tail of unusually poor high-latency measurements.

Week	Region	Carrier	Tech	Metric	Mean	Median proxy	Gap (%)
2026-03-01	Sohag	Operator B	Multi-RAT	Download	85,102.58	22,048.46	285.98
2026-03-08	Sohag	Operator B	Multi-RAT	Download	77,237.58	25,142.46	207.20
2026-03-01	Port Said	Operator C	5G	Latency	25.25	16.44	53.57
2026-03-08	Cairo	Operator C	LTE	Latency	54.42	35.77	52.13

Table 14: Illustrative cases where mean and median diverge strongly.

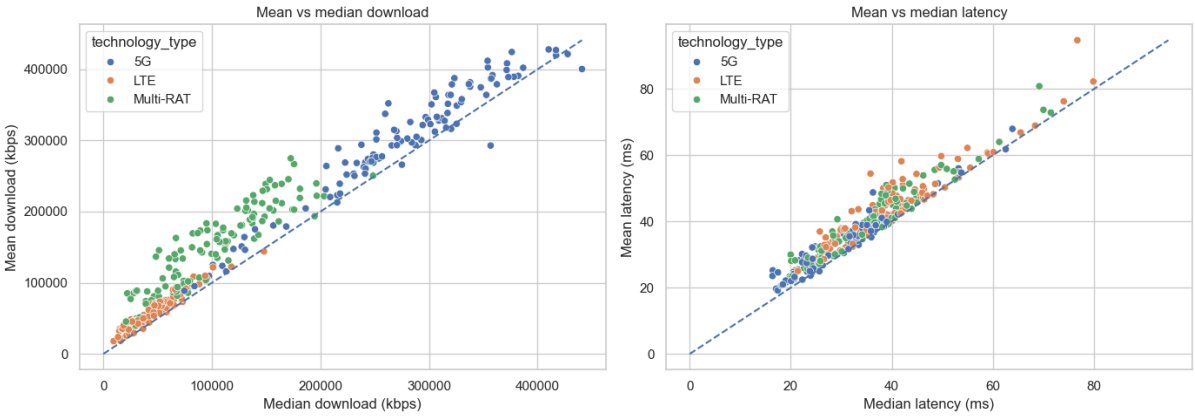

Figure 6: Mean vs median comparison for download and latency.

7 Limitations

The workflow is strong and appropriate for Project 6, but the following limitations should be remembered:

- The dataset contains only three weekly snapshots, so long-history temporal models such as rolling baselines, CUSUM, EWMA, or sequence models are not appropriate as primary methods.
- Median KPIs are **weighted proxies** derived from locality medians, not exact combined medians from raw speed-test records.
- The analysis is descriptive and anomaly-oriented. It identifies suspicious patterns but does not prove physical root cause.
- The threshold `sample_count_total >= 30` is a balanced choice for this dataset. It improves coverage compared with 50, but still removes the weakest region-carrier-technology-week rows.

- Complete peer-group filtering improves fairness for Q1 and Q3, but it excludes 13 rows that do not have all three carriers available in the same region-week-technology peer group.

8 Conclusion

The final corrected workflow produced a complete and explainable Project 6 solution. The most important conclusions are:

- **Operator C** is the main consistent underperformer relative to regional peers. Operator B is slightly below the weighted peer baseline but is much less severe and nearly neutral by the final week.
- The largest direct download drops were concentrated in **Assiut**, **Monufia**, **Gharbia**, **Minya**, and **Alexandria**.
- The strongest combined week-over-week anomalies were led by **Dakahlia** and **Port Said**.
- The strongest peer anomalies were repeatedly driven by **Operator C on LTE**, especially in Assiut, Ismailia, Giza, Alexandria, and Gharbia.
- **5G clearly outperformed LTE** across all region-week comparisons and all matched carrier-level comparisons.
- **Port Said** was the most unstable region by rank movement, while **Sohag** remained the weakest but stable bottom-ranked region.

9 Changes Compared with the First Report Version

The final report differs from the first version in the following important ways:

1. **Date parsing was corrected.** The earlier two-week issue was caused by forcing day-first parsing on ISO-formatted dates. The final notebooks preserve all three weeks.
2. **The sample threshold changed from 50 to 30.** This increased the final cleaned table from 335 to 343 rows and raised complete peer-comparison coverage from 312 to 330 rows.
3. **The peer anomaly score now exactly matches the slide formula.** The score is no longer a percentile-rank score; it is the sum of download underperformance and latency overperformance in bad percentage-gap points.
4. **Complete peer-group filtering was added.** Q1 and Q3 now use only groups

where all three carriers are present.

5. **Q2 was split into direct KPI changes and combined anomaly score.** The report now distinguishes largest download drops, largest latency rises, and combined week-over-week anomaly flags.
6. **The carrier report now separates worst-by-download from worst-by-peer-score.** This avoids mixing absolute weakness with relative anomaly severity.
7. **The 5G-vs-LTE section now includes a matched-carrier validation.** This confirms that 5G underperformed LTE in zero matched carrier-region-week cases.
8. **All tables were refreshed from the final generated CSV outputs.** The report tables now match the corrected preprocessing and analysis notebooks.

Appendix: Generated Files

The report was built alongside the following main outputs:

- `weekly_locality_only.csv`
- `weekly_country_only.csv`
- `weekly_region_carrier_tech.csv`
- `project6_carrier_report.csv`
- `project6_peer_anomaly_log.csv`
- `project6_change_anomaly_log.csv`
- `project6_anomaly_log.csv`
- `project6_region_ranking.csv`
- `project6_rank_stability.csv`
- `project6_fiveg_vs_lte.csv`
- `project6_fiveg_vs_lte_matched_carrier.csv`
- `project6_mean_median_examples.csv`